

Final Project 1: Mind Track - Brain task app

A production-ready React Single Page Application (SPA) containerized with Nginx and deployed automatically to an **Amazon Elastic Kubernetes Service (EKS)** cluster using an **AWS CodePipeline & CodeBuild** CI/CD pipeline.

Project Structure: <https://github.com/kousick-devaraj/Brain-tasks-app.git>

Architecture Diagram

[GitHub Repo] —(Push to main)—> [AWS CodePipeline] | ▼ [AWS CodeBuild]
|
| ————|————— ▼ ▼ [Build Image] [Deploy Manifests] | | ▼ ▼ [AWS ECR]
[Amazon EKS] | ▼ [AWS LoadBalancer] | ▼ [End User (Port 3000)]

Tech Stack

- **Frontend:** React (Pre-compiled dist static assets)
- **Web Server:** Nginx (alpine)
- **Containerization:** Docker
- **Registry:** AWS ECR | Storing container images
- **Cloud Provider:** AWS (VPC, EC2, EKS)
- **CI/CD Orchestration:** AWS CodeBuild & CodePipeline | Automated build & deployment
- **Container Orchestration:** AWS EKS (Kubernetes)
- **Monitoring:** Prometheus, Node Exporter & Grafana

1. Local Setup & Dockerization

- Cloning the source repository

```
MINGW64:/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1
$ git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 9 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 7.28 MiB/s, done.

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1
$ ls -lrt
total 0
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 Brain-Tasks-App/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1
$ cd Brain-Tasks-App/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 4
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

- Creating Dockerfile

The application uses a lightweight nginx:alpine image from AWS ECR Public to serve static assets from the dist/ directory on port 80.

```
MINGW64:/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 6
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 368 Sep 11 15:54 Dockerfile_backup
-rw-r--r-- 1 Kousick Devaraj 197121 198 Sep 11 15:56 Dockerfile

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ cat Dockerfile
FROM nginx:alpine

# Copy pre-compiled dist folder to Nginx standard HTML root
COPY dist /usr/share/nginx/html

# Nginx listens on default HTTP port 80
EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Building local image:

Build local image

docker build -t brain-tasks-app:local .

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker build -t brain-tasks-app:local .
[+] Building 19.0s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 237B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 318.06kB
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
=> => resolve docker.io/library/nginx:alpine@sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
=> sha256:8f924cf5086c7b5a67e8b3e03b22851c1de8e11b0433c21904ef340d89e5014e 1.40kB / 1.40kB
=> sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b23fa5e787232908e560673a66b553 20.52MB / 20.52MB
=> sha256:51900a10fb9cfad93aa74360e5f938efcbe6d3396a380867b7561c84e5cd2daf 1.21kB / 1.21kB
=> sha256:bc98d76756163f687c200ed6114df3904623bbb5e7698b420efaf8ef3c81dfcf 404B / 404B
=> sha256:58c524ea09ced2269cb14533d95b80de9fea9532abbf171585d8ace413893f2 956B / 956B
=> sha256:a17dd138f4591521064c208da6ef07f487ebc3e5b350989cdec2017b3e5208e3 628B / 628B
=> sha256:850bf2dceff7924b238f0b7cbaf7b456e5206453b784fe6b4dfc949add274a 4.40MB / 4.40MB
=> extracting sha256:850bf2dceff7924b238f0b7cbaf7b456e5206453b784fe6b4dfc949add274a 0.45s
=> extracting sha256:58c524ea09ced2269cb14533d95b80de9fea9532abbf171585d8ace413893f2 0.05s
=> extracting sha256:58c524ea09ced2269cb14533d95b80de9fea9532abbf171585d8ace413893f2 0.05s
=> extracting sha256:bc98d76756163f687c200ed6114df3904623bbb5e7698b420efaf8ef3c81dfcf 0.05s
=> extracting sha256:51900a10fb9cfad93aa74360e5f938efcbe6d3396a380867b7561c84e5cd2daf 0.05s
=> extracting sha256:8f924cf5086c7b5a67e8b3e03b22851c1de8e11b0433c21904ef340d89e5014e 0.05s
=> extracting sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b23fa5e787232908e560673a66b553 0.65s
=> [2/2] COPY dist /usr/share/nginx/html
=> exporting to image
=> exporting layers
=> exporting manifest sha256:ab443c01d223c548d42f121e49813e505f61b697e5bf3a9b29bd3a491b384855 0.05s
=> exporting config sha256:8b72771fcf5d6e2e9fef7ac05768da40ee82fb9e5a30bd25c46df65d7ad6b 0.05s
=> exporting attestation manifest sha256:9d54f3227099b9f59cf21b1e2e9f23eed156d2d727df314e54e0eb49cafb6a7 0.05s
=> exporting manifest list sha256:c324674683c0ccdcfa0f6a7bd531542be7d5d5ac258b30560e20fc665ef857f1 0.05s
=> naming to docker.io/library/brain-tasks-app:local
=> unpacking to docker.io/library/brain-tasks-app:local 0.05s

view build details: docker-desktop://dashboard/build/docker-linux/desktop-linux/niymhmvow7wy9k6elwrhnttt
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

Run container mapping host port 3000 to Nginx port 80

docker run -d -p 3000:80 --name brain-tasks-test brain-tasks-app:local

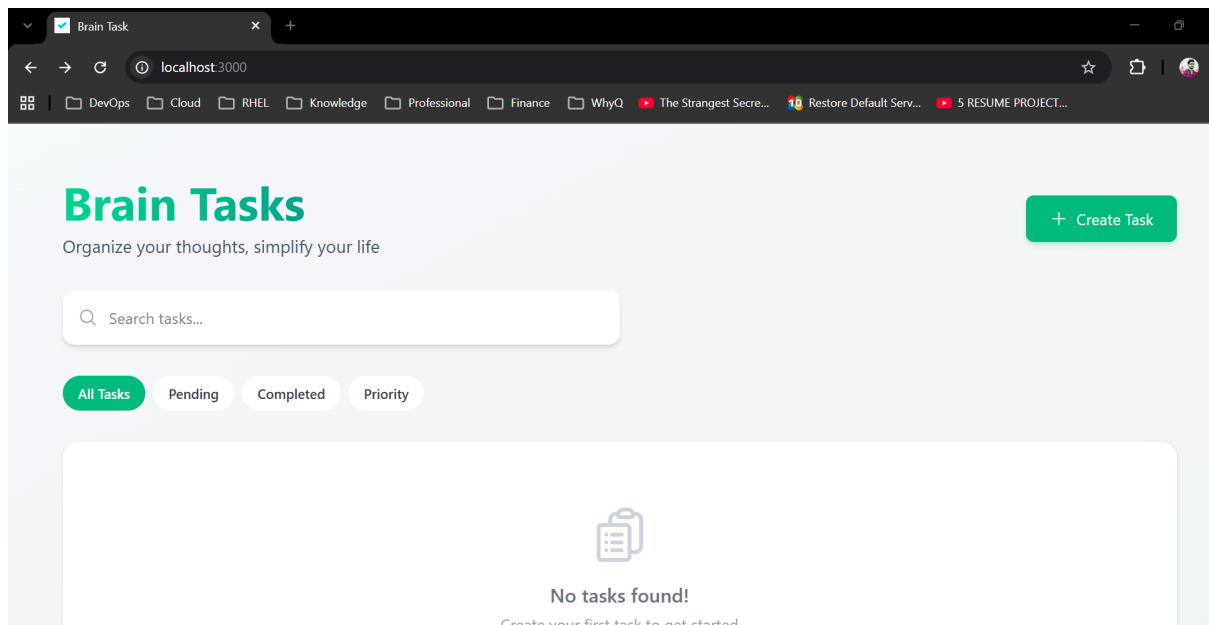
```
MINGW64:/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker run -d -p 3000:80 --name brain-tasks-test brain-tasks-app:local
114c68b5f1056fb68b685a7103982578836d6382b41b4df8cd097b9313538c53

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS       PORTS                               NAMES
114c68b5f105   brain-tasks-app:local   "/docker-entrypoint.…"   6 seconds ago   Up 6 seconds   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   brain-tasks-test

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

The app is validated on a local machine and it is working using a container.



Create your own GitHub Repository & Push the Codebase

```
MINGW64/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 6
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 368 Sep 11 15:54 Dockerfile_backup
-rw-r--r-- 1 Kousick Devaraj 197121 198 Sep 11 15:56 Dockerfile

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git remote add origin https://github.com/kousick-devaraj/Brain-tasks-app.git

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git branch -M main

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile
        Dockerfile_backup

nothing added to commit but untracked files present (use "git add" to track)

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git add Dockerfile
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Dockerfile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile_backup

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git commit -m "Initial commit with custom Dockerfile to create image for Brain task app"
[main 55ba793] Initial commit with custom Dockerfile to create image for Brain task app
1 file changed, 9 insertions(+)
create mode 100644 Dockerfile

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git push -u origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 8 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (14/14), 102.35 KiB | 51.18 MiB/s, done.
Total 14 (delta 0), reused 11 (delta 0), pack-reused 0 (from 0)
To https://github.com/kousick-devaraj/Brain-tasks-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ |
```

2. AWS ECR Repository Creation

#Create a private container registry to store application images:

aws ecr create-repository --repository-name brain-tasks-app --region us-east-1

```
MINGW64:/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ aws ecr create-repository --repository-name brain-tasks-app --region us-east-1
{
  "repository": {
    "repositoryArn": "arn:aws:ecr:us-east-1:786352305260:repository/brain-tasks-app",
    "registryId": "786352305260",
    "repositoryName": "brain-tasks-app",
    "repositoryUri": "786352305260.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app",
    "createdAt": "2026-09-11T16:59:28.153000+05:30",
    "imageTagMutability": "MUTABLE",
    "imageScanningConfiguration": {
      "scanOnPush": false
    },
    "encryptionConfiguration": {
      "encryptionType": "AES256"
    }
  }
}

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
114c68b5f105   brain-tasks-app:local   "/docker-entrypoint..." 59 minutes ago Up 59 minutes 0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   brain-tasks-test

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker tag brain-tasks-app:local 786352305260.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1.0.0

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
114c68b5f105   brain-tasks-app:local   "/docker-entrypoint..." 59 minutes ago Up 59 minutes 0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   brain-tasks-test

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ docker push 786352305260.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1.0.0
The push refers to repository [786352305260.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
0fb44b85ff94: Pushed
44136fa355b3: Pushed
850bf2dcecff: Pushed
55afalecc21d: Pushed
af7dd138f459: Pushed
58c524ea09ce: Pushed
bc98d7675616: Pushed
51900e10fb9c: Pushed
6636b9fc203c: Pushed
8f924cf5086c: Pushed
dfc8b3982a8a: Pushed
v1.0.0: digest: sha256:c324674683c0ccdafe0f6a7bd551542be7d5d5ac258b30560e20fc6656f857f1 size: 856
```

The screenshot shows the AWS Management Console interface for the 'brain-tasks-app' repository. The left sidebar shows the navigation menu with 'Amazon Elastic Container Registry' selected. The main content area shows the repository details, including the 'Images' tab which lists three images. The first image, 'v1.0.0', is an 'Image Index' with a size of 28.88 MB and a digest of 'sha256:c324674683c0ccdafe0f6a7bd551542be7d5d5ac258b30560e20fc6656f857f1'. The other two images are blank and labeled 'Other' and 'Image'.

3. AWS EKS Cluster Provisioning

Provision an EKS cluster with two worker nodes:

Bash

**eksctl create cluster **

**--name brain-tasks-cluster **

**--region us-east-1 **

**--nodegroup-name standard-workers **

**--node-type t3.medium **

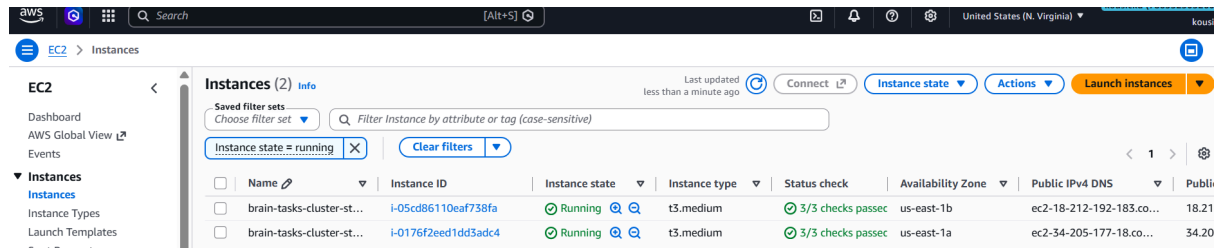
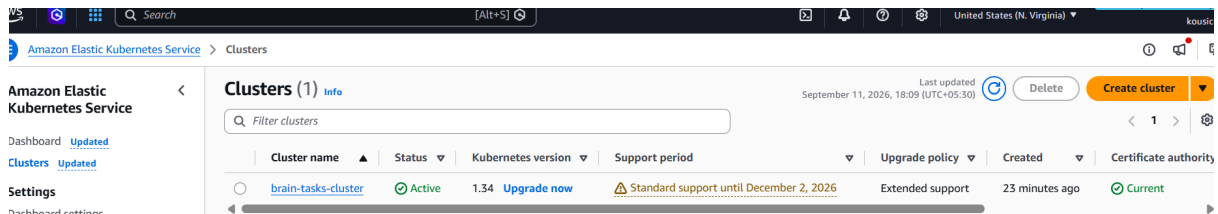
**--nodes 2 **

--managed

Connect local kubectl context

aws eks update-kubeconfig --region us-east-1 --name brain-tasks-cluster

```
MINGW64\c:\Users\Kousick Devaraj\Desktop\GuviProjects\Project1\Brain-Tasks-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ eksctl create cluster --name brain-tasks-cluster --region us-east-1 --nodegroup-name standard-workers --node-type t3.medium --nodes 2 --managed
2026-09-11 17:46:02 [i] eksctl version 0.230.0
2026-09-11 17:46:02 [i] using region us-east-1
2026-09-11 17:46:03 [i] setting availability zones to [us-east-1a us-east-1b]
2026-09-11 17:46:03 [i] subnets for us-east-1a - public:192.168.0.0/19 private:192.168.64.0/19
2026-09-11 17:46:03 [i] subnets for us-east-1b - public:192.168.32.0/19 private:192.168.96.0/19
2026-09-11 17:46:03 [i] nodegroup "standard-workers" will use "" [AmazonLinux2023/1.34]
2026-09-11 17:46:03 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer
be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto
mode/
2026-09-11 17:46:03 [i] using Kubernetes version 1.34
2026-09-11 17:46:03 [i] creating EKS cluster "brain-tasks-cluster" in "us-east-1" region with managed nodes
2026-09-11 17:46:03 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-09-11 17:46:03 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=brain-tasks-cluster'
2026-09-11 17:46:03 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "brain-tasks-cluster" in "us-east-1"
2026-09-11 17:46:03 [i] Cloudwatch logging will not be enabled for cluster "brain-tasks-cluster" in "us-east-1"
2026-09-11 17:46:03 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=br
ain-tasks-cluster'
2026-09-11 17:46:03 [i] default add-ons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS add-ons
2026-09-11 17:46:03 [i] 2 sequential tasks: { create cluster control plane "brain-tasks-cluster",
2 sequential sub-tasks: {
2 sequential sub-tasks: {
1 task: { create add-ons },
wait for control plane to become ready,
},
create managed nodegroup "standard-workers",
}
}
2026-09-11 17:46:03 [i] building cluster stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:46:04 [i] deploying stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:46:34 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:47:05 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:48:05 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:48:07 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:50:08 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:51:09 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:52:10 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:53:11 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:54:12 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:55:14 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:56:15 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:57:16 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:58:17 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-09-11 17:58:23 [i] recommended policies were found for "vpc-cni" add-on, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions;
the recommended way to provide IAM permissions for "vpc-cni" add-on is via pod identity associations; after add-on creation is completed, add all recommended policies to the
2026-09-11 17:58:23 [i] recommended policies were found for "vpc-cni" add-on, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions;
the recommended way to provide IAM permissions for "vpc-cni" add-on is via pod identity associations; after add-on creation is completed, add all recommended policies to the
2026-09-11 17:58:23 [i] creating add-on: vpc-cni
2026-09-11 17:58:23 [i] successfully created add-on: vpc-cni
2026-09-11 17:58:24 [i] creating add-on: kube-proxy
2026-09-11 17:58:25 [i] successfully created add-on: kube-proxy
2026-09-11 17:58:25 [i] creating add-on: coredns
2026-09-11 17:58:26 [i] successfully created add-on: coredns
2026-09-11 18:00:32 [i] building managed nodegroup stack "eksctl-brain-tasks-cluster-nodegroup-standard-workers"
2026-09-11 18:00:34 [i] deploying stack "eksctl-brain-tasks-cluster-nodegroup-standard-workers"
2026-09-11 18:00:34 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-standard-workers"
2026-09-11 18:01:06 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-standard-workers"
2026-09-11 18:02:05 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-standard-workers"
2026-09-11 18:03:03 [i] waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-standard-workers"
2026-09-11 18:03:03 [i] waiting for the control plane to become ready
2026-09-11 18:03:07 [i] saved kubeconfig as "c:\Users\Kousick Devaraj\kube\config"
2026-09-11 18:03:07 [i] no tasks
2026-09-11 18:03:07 [i] all EKS cluster resources for "brain-tasks-cluster" have been created
2026-09-11 18:03:09 [i] nodegroup "standard-workers" has 2 node(s)
2026-09-11 18:03:09 [i] node "ip-192-168-17-214.ec2.internal" is ready
2026-09-11 18:03:09 [i] node "ip-192-168-41-22.ec2.internal" is ready
2026-09-11 18:03:09 [i] waiting for at least 2 node(s) to become ready in "standard-workers"
2026-09-11 18:03:09 [i] nodegroup "standard-workers" has 2 node(s)
2026-09-11 18:03:09 [i] node "ip-192-168-17-214.ec2.internal" is ready
2026-09-11 18:03:09 [i] node "ip-192-168-41-22.ec2.internal" is ready
2026-09-11 18:03:09 [i] created 1 managed nodegroup(s) in cluster "brain-tasks-cluster"
2026-09-11 18:03:11 [i] creating add-on: metrics-server
2026-09-11 18:03:11 [i] successfully created add-on: metrics-server
2026-09-11 18:03:16 [i] kubectl command should work with "c:\Users\Kousick Devaraj\kube\config", try 'kubectl get nodes'
2026-09-11 18:03:16 [i] EKS cluster "brain-tasks-cluster" in "us-east-1" region is ready
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```



4. Kubernetes Manifests

- Creating the k8s deployment manifest and services.

```

MINGW64:/c:/Users/Kousick Devaraj/Desktop/GuviProjects/Project1/Brain-Tasks-App/k8s
$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
  labels:
    app: brain-tasks-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks-app
  template:
    metadata:
      labels:
        app: brain-tasks-app
    spec:
      containers:
        - name: brain-tasks-container
          image: 786352305260.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
          ports:
            - containerPort: 80
          resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
            limits:
              cpu: "250m"
              memory: "256Mi"

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App/k8s (main)
$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  ports:
    - name: http
      port: 3000
      targetPort: 80
      protocol: TCP
  selector:
    app: brain-tasks-app

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App/k8s (main)

```

Pushing data to ECR :

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ ls -lrt
total 10
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 11 15:38 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 15:38 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 368 Sep 11 15:54 Dockerfile_backup
-rw-r--r-- 1 Kousick Devaraj 197121 198 Sep 11 15:56 Dockerfile
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 11 18:16 k8s/
-rw-r--r-- 1 Kousick Devaraj 197121 1533 Sep 11 18:29 buildspec.yml

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile_backup
    buildspec.yml
    k8s/

nothing added to commit but untracked files present (use "git add" to track)

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git add k8s/ buildspec.yml
warning: in the working copy of 'buildspec.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/service.yaml', LF will be replaced by CRLF the next time Git touches it

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   buildspec.yml
    new file:   k8s/deployment.yaml
    new file:   k8s/service.yaml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile_backup

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
```

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.52 KiB | 310.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kousick-devaraj/Brain-tasks-app.git
  55ba793..c122e4e  main -> main

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project1/Brain-Tasks-App (main)
$
```

ECR Repo and image

Amazon Elastic Container Registry

Private registry

Repositories

brain-tasks-app

Features & Settings

Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

brain-tasks-app

Summary Images Lifecycle policy Permissions Repository tags

Images (3) Info

Filter active images

Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
v1.0.0	Image Index	September 11, 2026, 17:01:13 (UTC+05:5)	28.88	sha256:c324674683c0ccd...	-
-	Other	September 11, 2026, 17:01:12 (UTC+05:5)	0.00	sha256:9d54f3227099bff...	-
-	Image	September 11, 2026, 17:01:12 (UTC+05:5)	28.88	sha256:ab443c01d223c54...	-

Buttons: Delete, Copy URI, Details, Scan, View push commands

Cluster info

The screenshot shows the Amazon Elastic Kubernetes Service (EKS) console for the cluster 'brain-tasks-cluster'. The left sidebar contains navigation links for Dashboard, Settings, Amazon EKS Anywhere, and Related services. The main content area displays cluster details:

- Cluster info:** Status is Active, Kubernetes version is 1.34, Support period is Standard support until December 2, 2026, and Provider is EKS.
- Cluster health:** 0 issues.
- Upgrade insights:** 4 insights.
- Node health issues:** 0 issues.
- Capability issues:** 0 issues.

Below the summary, the 'Details' section provides more information:

- API server endpoint:** <https://9727f12b511e0595ccbd32bf97a0b0ff.gr7.us-east-1.eks.amazonaws.com>
- OpenID Connect provider URL:** <https://oidc.eks.us-east-1.amazonaws.com/id/9727f12b511e0595ccbd32bf97a0b0ff>
- Certificate authority:** [View in IAM](#)
- Cluster IAM role ARN:** [arn:aws:iam::786352305260:role/eksctl-brain-tasks-cluster-cluster-Ser viceRole-Lq3OK4T14gJO](#)
- Created:** 6 hours ago
- Cluster ARN:** [arn:aws:eks:us-east-1:786352305260:cluster/brain-tasks-cluster](#)
- Platform version:** eks.32

5. IAM & EKS Authorization for CodeBuild

To authorize AWS CodeBuild to make deployment changes to EKS, map its IAM Role (CodeBuildEKSDeployRole) inside EKS:

Bash

```
eksctl create iamidentitymapping \
  --cluster brain-tasks-cluster \
  --region us-east-1 \
  --arn arn:aws:iam::<YOUR_ACCOUNT_ID>:role/CodeBuildEKSDeployRole \
  --group system:masters \
  --username build
```

The screenshot shows the AWS IAM console for the role 'CodeBuildEKSDeployRole'. The 'Permissions' tab is selected, displaying a list of 9 policies attached to the role:

Policy name	Type	Attached entities
AmazonEC2ContainerRegistryPowerUser	AWS managed	2
AmazonEKSClusterPolicy	AWS managed	4
CloudWatchLogsFullAccess	AWS managed	2
CodeBuildBasePolicy-CodeBuildEKSDeployRole-us-east-1	Customer managed	1
CodeBuildCloudWatchLogsPolicy	Customer inline	0
CodeBuildCloudWatchLogsPolicy-MindTrack-us-east-1	Customer managed	1
CodeBuildCodeConnectionsSourceCredentialsPolicy-MindTrack-us-east-1	Customer managed	1
CodeBuildEKSDeployPolicy	Customer inline	0
CodeBuildVpcPolicy-MindTrack-us-east-1	Customer managed	1

6. Automated CI/CD Pipeline Configuration

The pipeline triggers automatically on every git push to main.

1. **Source:** GitHub via AWS CodeConnections.
2. **Build:** AWS CodeBuild compiles Docker images, tags them with commit hashes, pushes to ECR, and applies Kubernetes manifests (kubectl apply).
3. **Deploy:** Managed inside CodeBuild using cluster credentials.

The screenshot shows the AWS CodeBuild console for the 'MindTrack' project. The left sidebar contains navigation links for Developer Tools, CodeBuild, and Build projects. The main area displays the project configuration and a table of build history.

Configuration:

- Source provider: GitHub
- Primary repository: kousick-devaraj/Brain-tasks-app 1.0
- Artifacts upload location: -
- Service role: arn:aws:iam::786352305260:role/service-role/CodeBuildEKSDeployRole
- Public builds: Disabled

Build history:

Build run	Status	Build number	Source version	Submitter	Duration	Completed
MindTrack:6560f8-3319-4d5a-9e62-a35a4667a465	Succeeded	18	arn:aws:s3::codepipeline-us-east-1-d0aa71b62a42-41be-b555-d4b24b0a0c58/brain-tasks-pipeline/SourceArti/4RlieAw	codepipeline/brain-tasks-pipeline	31 seconds	23 minutes ago
MindTrack:3703e8dc-07eb-49a9-9e4f-ed3f5c990b0e	Failed	17	arn:aws:s3::codepipeline-us-east-1-d0aa71b62a42-41be-b555-d4b24b0a0c58/brain-tasks-pipeline/SourceArti/QbVVD6g	codepipeline/brain-tasks-pipeline	38 seconds	27 minutes ago

Pipeline

The screenshot shows the AWS CodePipeline console for the 'brain-tasks-pipeline'. A green success banner at the top states: 'Congratulations! The pipeline brain-tasks-pipeline has been created.' The pipeline diagram shows two stages: 'Source' and 'Build'. The 'Source' stage is completed, and the 'Build' stage is in progress.

brain-tasks-pipeline

Source Stage:

- 13607a79-0265-4495-bf9c-63b30f3d451d
- All actions succeeded.
- Source: GitHub (via GitHub App) Just now
- c122e4e9 Source: Adding K8s manifests and CodeBuild specification

Build Stage:

- 13607a79-0265-4495-bf9c-63b30f3d451d
- In progress: 1
- Build: AWS CodeBuild Just now
- c122e4e9 Source: Adding K8s manifests and CodeBuild specification

The screenshot shows the AWS CodePipeline console for the 'brain-tasks-pipeline'. A green success banner at the top states: 'The most recent change will re-run through the pipeline. It might take a few moments for the status of the run to show in the pipeline view.' The pipeline diagram shows two stages: 'Source' and 'Build'. Both stages are completed.

brain-tasks-pipeline

Source Stage:

- 67458b29-9cc6-40f5-aa7f-fb6442c984ed
- All actions succeeded.
- Source: GitHub (via GitHub App) 10 minutes ago
- efa7fec2 Source: removed conflict markers

Build Stage:

- 67458b29-9cc6-40f5-aa7f-fb6442c984ed
- All actions succeeded.
- Build: AWS CodeBuild 15 minutes ago
- efa7fec2 Source: removed conflict markers

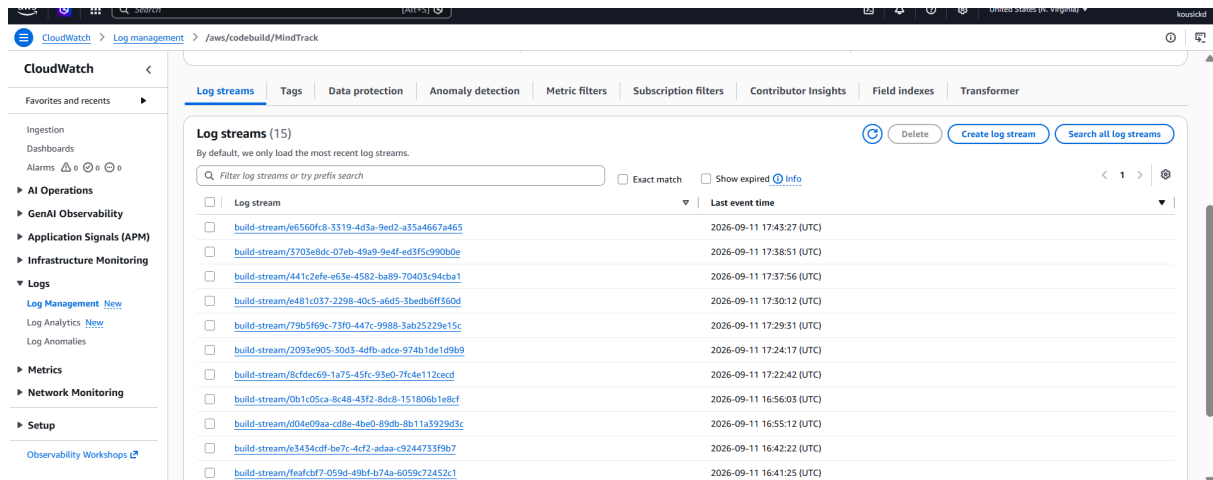
7. Monitoring & Operational Logs

Build execution and container deployment events are streamed to CloudWatch Logs under the `/aws/codebuild/brain-tasks-build` log group.

The screenshot shows the AWS CodeBuild console interface. On the left is a navigation pane with options like Source, Artifacts, Build, and Build projects. The main area displays the build details for a specific build. The build status is 'Succeeded'. Key information includes the build ARN, start time (Sep 11, 2026 11:13 PM UTC+5:30), end time (Sep 11, 2026 11:13 PM UTC+5:30), and build number 18. Below this, there are tabs for Build logs, Phase details, Reports, Environment variables, Build details, and Resource utilization. The 'Build logs' tab is active, showing a list of log entries. The first entry is '[Container] 2026/09/11 17:43:22.078953 Running on CodeBuild On-demand'. The second entry is '[Container] 2026/09/11 17:43:22.078977 Waiting for agent ping'. The third entry is '[Container] 2026/09/11 17:43:22.765011 Waiting for DOWNLOAD_SOURCE'. The fourth entry is '[Container] 2026/09/11 17:43:23.711237 Phase is DOWNLOAD_SOURCE'. The fifth entry is '[Container] 2026/09/11 17:43:23.711608 CODEBUILD_SRC_DIR=/codebuild/output/src1037853588/src'. The sixth entry is '[Container] 2026/09/11 17:43:23.713514 YAML location is /codebuild/output/src1037853588/src/buildspec.yml'. The seventh entry is '[Container] 2026/09/11 17:43:23.717094 Setting HTTP client timeout to higher timeout for S3 source'. The eighth entry is '[Container] 2026/09/11 17:43:23.717242 Processing environment variables'. The ninth entry is '[Container] 2026/09/11 17:43:24.032506 No runtime version selected in buildspec'. The tenth entry is '[Container] 2026/09/11 17:43:24.059989 Moving to directory /codebuild/output/src1037853588/src'. The eleventh entry is '[Container] 2026/09/11 17:43:24.060814 Cache is not defined in the buildspec'. The twelfth entry is '[Container] 2026/09/11 17:43:24.228095 Skip cache due to: no paths specified to be cached'.

This screenshot continues the log output from the previous one. The first entry is '[Container] 2026/09/11 17:43:41.535963 Running command echo Updating kubeconfig for EKS...'. The second entry is '[Container] 2026/09/11 17:43:41.543350 Running command aws eks update-kubeconfig --region \$AWS_REGION --name \$EKS_CLUSTER_NAME'. The third entry is '[Container] 2026/09/11 17:43:42.512721 Running command echo Deploying manifests to EKS...'. The fourth entry is '[Container] 2026/09/11 17:43:42.520053 Running command sed -i 's|<ACCOUNT_ID>|<ACCOUNT_ID>|' /etc/ecs/ecs-credentials.json'. The fifth entry is '[Container] 2026/09/11 17:43:42.563804 Running command kubectl apply -f k8s/deployment.yaml'. The sixth entry is '[Container] 2026/09/11 17:43:43.489510 Running command kubectl apply -f k8s/service.yaml'. The seventh entry is '[Container] 2026/09/11 17:43:44.485381 Running command kubectl rollout status deployment/brain-tasks-deployment'. The eighth entry is '[Container] 2026/09/11 17:43:47.588686 Phase complete: POST_BUILD State: SUCCEEDED'. The ninth entry is '[Container] 2026/09/11 17:43:47.588626 Phase context status code: Message:'. The tenth entry is '[Container] 2026/09/11 17:43:47.587263 Set report auto-discover timeout to 5 seconds'. The eleventh entry is '[Container] 2026/09/11 17:43:47.587261 Expanding base directory path: .'. The twelfth entry is '[Container] 2026/09/11 17:43:47.586813 Assembling file list'. The thirteenth entry is '[Container] 2026/09/11 17:43:47.586649 Expanding .'. The fourteenth entry is '[Container] 2026/09/11 17:43:47.571993 Expanding file paths for base directory .'. The fifteenth entry is '[Container] 2026/09/11 17:43:47.572003 Assembling file list'. The sixteenth entry is '[Container] 2026/09/11 17:43:47.575573 No matching auto-discover report paths found'. The seventeenth entry is '[Container] 2026/09/11 17:43:47.575589 Report auto-discover file discovery took 0.000326 seconds'. The eighteenth entry is '[Container] 2026/09/11 17:43:47.575632 Phase complete: UPLOAD_ARTIFACTS State: SUCCEEDED'. The nineteenth entry is '[Container] 2026/09/11 17:43:47.575638 Phase context status code: Message:'.

Cloud watch logs



Verification & Live Application

To fetch the dynamic AWS LoadBalancer URL:

```
kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Downloads
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-7c98b7bc6b-mj8tr  1/1     Running   0           4m27s
brain-tasks-deployment-7c98b7bc6b-qh6d1  1/1     Running   0           4m26s

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Downloads
$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/brain-tasks-deployment-7c98b7bc6b-mj8tr  1/1     Running   0           5m7s
pod/brain-tasks-deployment-7c98b7bc6b-qh6d1  1/1     Running   0           5m6s

NAME                                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
service/brain-tasks-service         LoadBalancer  10.100.206.51 a87010d6dcd614b718c9e322d9740e23-1814529202.us-east-1.elb.amazonaws.com 3000:30699/TCP 5m7s
service/kubernetes                  ClusterIP      10.100.0.1    <none>        <none>           5h25m

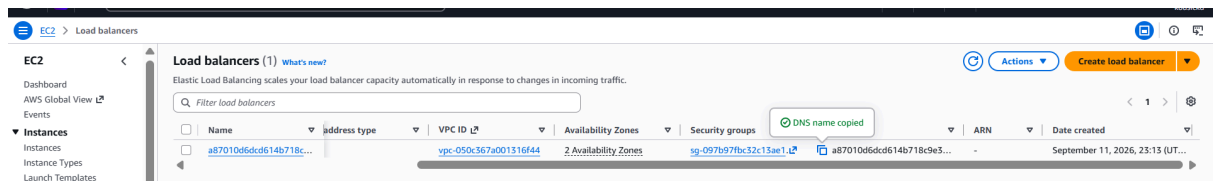
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/brain-tasks-deployment  2/2     2             2           5m8s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/brain-tasks-deployment-7c98b7bc6b  2         2         2       5m9s

kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Downloads
```

LB url -

<http://a87010d6dcd614b718c9e322d9740e23-1814529202.us-east-1.elb.amazonaws.com:3000/>



Validating output using LoadBalancer URL of the k8s cluster.

Brain Tasks

Organize your thoughts, simplify your life

+ Create Task

Search tasks...

- All Tasks Pending Completed Priority



No tasks found!
Create your first task to get started